17/9/26

EXP- NO: 06

HAMMING CODE

Aim: Write program to Implement error correction and correction using hamming code concept make not to Input data stream & verify correction

CODE:

```
def calc_redundant_bits(m):
    for r in range(m+1):
        if (2**r)>=(m+r+1):
            return r

def main():
    data=input("Enter data min:").strip()
    m=len(data)
    r=calc_redundant_bits(m)
    n=m+r
    print(f"number of r needed:{r}")
    print(f"Total frame length:{n}")
    code=['0']+(n+1)
    j=0
    for i in range(1,n+1):
        if(i&(i-1))!=0:
            code[i]=data[(j+1)]
            j+=1
    for i in range(r):
        parity-pos=2**i;
        var=0
        for pos in range(1,n+1):
            if(pos&parity-pos)!=0 and
                pos!=parity-pos
                if code[pos]=='1':
                    val^=1

        val[parity-pos]=str(val)
```

```
transmitted_str = " ".join(code[1:][::-1])
print(f"Data to be transmitted:
      {transmitted transmitted_str}")
change = input("\n Transmitted bit
      change").strip().upper()
if change == "YES":
    received_str = input(f"Enter corrupted
data").strip()
    if len(received_str) != n:
        print(f"Error: received data be
exactly {n} bits ng!")
        return
    received_code = ['0'] + list(received_str
                                 [::-1])
else: else:
    received_code = list(code)
error_position = 0
for i in range(r):
    parity_pos = 2**i;
    val = 0
    for pos in range(1,n+1):
        if (pos & parity_pos):= 0
            if received_code[pos] == '1':
                val n=1
    if val != 0
        error_position += parity pos
if error_position == 0:
    print("\n Verification Result: No
error detected")
else:
```

```
        print("\n Verification Result: Error
            detected!")
        print(f"Wrong bit at position: {error_
            position}")
        received_code[error_position] = '0' if
                received_code
                [error_position] == '1' else '1'

        corrected_str = "".join(received_code[1:]
                        [::-1])
        print(f"Corrected data stream: {corrected_
            str}")

if __name__ == "__main__":
        main()
```

OUTPUT:

(1)

```
Enter data bits: 101100
Number of redundant bits (r) needed: 4
Total frame length (m+r): 10
Data to be transmitted: 1011100011
Transmitted bit change? (YES/NO): YES/NO: YES
Enter the corrupted data stream (10 bits):
                                1011101011

verification Result: Error detected!
Wrong bit is at position: 4
Corrected data stream: 1011100011
```

(2)

```
Error data bits: 1011
number of redundant bits (r) needed: 3
Total data frame length (m+r): 7
Data to be transmitted: 1010101
```

Transmitted bit change? (YES/NO): NO
Verification Result: no error detected
in transmission

Result: Thus the Hamming code was implemented successfully for error detection & correction